COVID-19 Data Analysis Report

Python cleaning, a Power BI dashboard, findings and recommendations

187 countries · snapshot of 27 July 2020 · Prepared by Hager Naser

DIRECT ANSWER · PART 1

Overall statistical analysis

Category	Total	Average	Maximum	Max country
Active	6,358,362	34,001.94	2,816,444	US
Confirmed	16,480,485	88,130.94	4,290,259	US
Deaths	654,036	3,497.52	148,011	US
Recovered	9,468,087	50,631.48	1,846,641	Brazil

Average = mean of all 187 countries

Not filtered by any selection, so it matches the dashboard cards.

The US holds the maximum in 3 of 4

Brazil leads Recovered; the US is second with 1,325,804.

Source: country_wise_latest · 187 countries · snapshot of 27 Jul 2020

DIRECT ANSWER · PART 2

Countries above the average

Category	Average	Countries above	Share of 187	Highest country
Active	34,002	24	12.8%	US · 2,816,444
Confirmed	88,131	25	13.4%	US · 4,290,259
Deaths	3,498	26	13.9%	US · 148,011
Recovered	50,631	28	15.0%	Brazil · 1,846,641

“Overactive” means “above average”

The requirements define no separate threshold, so a country is flagged whenever its value is greater than the average of that category. The next four slides list every flagged country; the full 187-country tables are in the dashboard.

Source: country_wise_latest · 187 countries · snapshot of 27 Jul 2020

DIRECT ANSWER · PART 2 · ACTIVE

Active: 24 countries above 34,002

24 of 187 countries (12.8%) together hold 89.3% of all active cases

#	Country	Active
1	US	2,816,444
2	Brazil	508,116
3	India	495,499
4	United Kingdom	254,427
5	Russia	201,097
6	South Africa	170,537
7	Colombia	117,163
8	France	108,928

#	Country	Active
9	Canada	107,514
10	Peru	98,752
11	Bangladesh	97,577
12	Spain	93,613
13	Argentina	91,782
14	Sweden	73,695
15	Philippines	53,649
16	Egypt	52,992

#	Country	Active
17	Mexico	47,657
18	Netherlands	47,064
19	Bolivia	47,056
20	Saudi Arabia	43,238
21	Ecuador	40,733
22	Belgium	39,154
23	Indonesia	37,292
24	Israel	36,378

Source: country_wise_latest · 187 countries · snapshot of 27 Jul 2020

DIRECT ANSWER · PART 2 · CONFIRMED

Confirmed: 25 countries above 88,131

25 of 187 countries (13.4%) together hold 85.6% of all confirmed cases

#	Country	Confirmed
1	US	4,290,259
2	Brazil	2,442,375
3	India	1,480,073
4	Russia	816,680
5	South Africa	452,529
6	Mexico	395,489
7	Peru	389,717
8	Chile	347,923
9	United Kingdom	301,708

#	Country	Confirmed
10	Iran	293,606
11	Pakistan	274,289
12	Spain	272,421
13	Saudi Arabia	268,934
14	Colombia	257,101
15	Italy	246,286
16	Turkey	227,019
17	Bangladesh	226,225

#	Country	Confirmed
18	France	220,352
19	Germany	207,112
20	Argentina	167,416
21	Canada	116,458
22	Iraq	112,585
23	Qatar	109,597
24	Indonesia	100,303
25	Egypt	92,482

Source: country_wise_latest · 187 countries · snapshot of 27 Jul 2020

DIRECT ANSWER · PART 2 · DEATHS

Deaths: 26 countries above 3,498

26 of 187 countries (13.9%) together hold 91.8% of all deaths

#	Country	Deaths
1	US	148,011
2	Brazil	87,618
3	United Kingdom	45,844
4	Mexico	44,022
5	Italy	35,112
6	India	33,408
7	France	30,212
8	Spain	28,432
9	Peru	18,418

#	Country	Deaths
10	Iran	15,912
11	Russia	13,334
12	Belgium	9,822
13	Chile	9,187
14	Germany	9,125
15	Canada	8,944
16	Colombia	8,777
17	South Africa	7,067
18	Netherlands	6,160

#	Country	Deaths
19	Pakistan	5,842
20	Sweden	5,700
21	Turkey	5,630
22	Ecuador	5,532
23	Indonesia	4,838
24	China	4,656
25	Egypt	4,652
26	Iraq	4,458

Source: country_wise_latest · 187 countries · snapshot of 27 Jul 2020

DIRECT ANSWER · PART 2 · RECOVERED

Recovered: 28 countries above 50,631

28 of 187 countries (15.0%) together hold 88.5% of all recovered cases

#	Country	Recovered
1	Brazil	1,846,641
2	US	1,325,804
3	India	951,166
4	Russia	602,249
5	Chile	319,954
6	Mexico	303,810
7	South Africa	274,925
8	Peru	272,547
9	Iran	255,144
10	Pakistan	241,026

#	Country	Recovered
11	Saudi Arabia	222,936
12	Turkey	210,469
13	Italy	198,593
14	Germany	190,314
15	Spain	150,376
16	Colombia	131,161
17	Bangladesh	125,683
18	Qatar	106,328
19	France	81,212

#	Country	Recovered
20	China	78,869
21	Iraq	77,144
22	Argentina	72,575
23	Belarus	60,492
24	Indonesia	58,173
25	Oman	57,028
26	Kuwait	55,057
27	Kazakhstan	54,404
28	UAE	52,510

Source: country_wise_latest · 187 countries · snapshot of 27 Jul 2020 · UAE = United Arab Emirates

METHOD

Data preparation in Python

STEP 1

Load

Read the country_wise_latest sheet: 187 countries, one row per country.

STEP 2

Clean

Dropped two stray columns (Unnamed: 15 and 16). Their only value, 4,290,259, is the US confirmed total.

STEP 3

Validate

No missing values, duplicates or negatives.

STEP 4

Prepare

Added an Above_Avg flag per category, saved each requirement as a table and exported the clean file to Power BI.

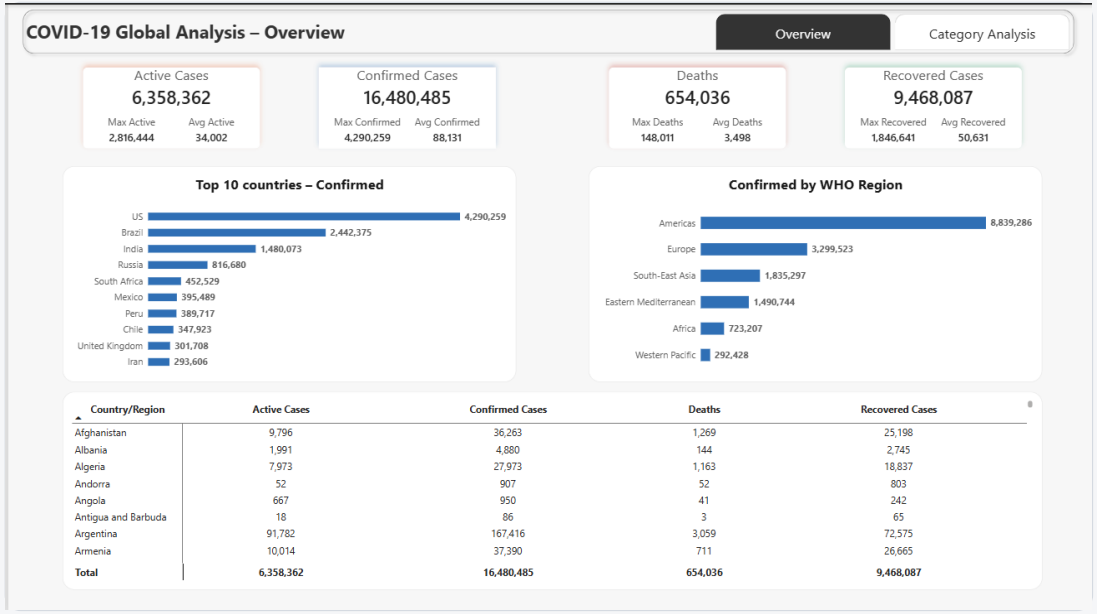
Country names were kept as in the source

For example Taiwan* and Congo (Kinshasa). Each country appears once, so no grouping was needed before comparing with the average.

Source: country_wise_latest · 187 countries · snapshot of 27 Jul 2020

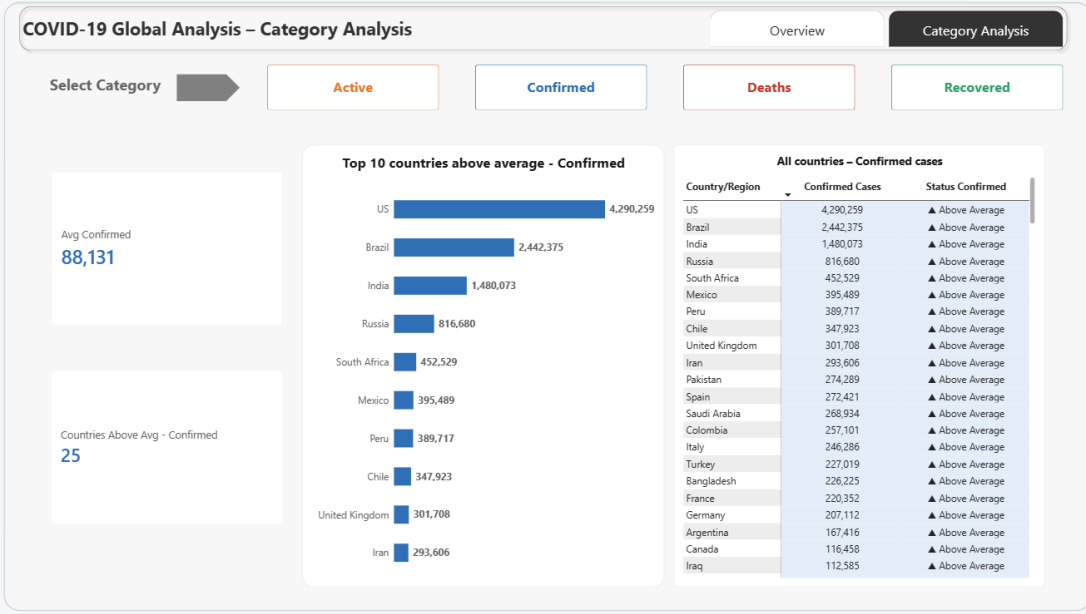
DASHBOARD

The Power BI dashboard



Page 1 · Overview

Part 1: totals, maximum and average



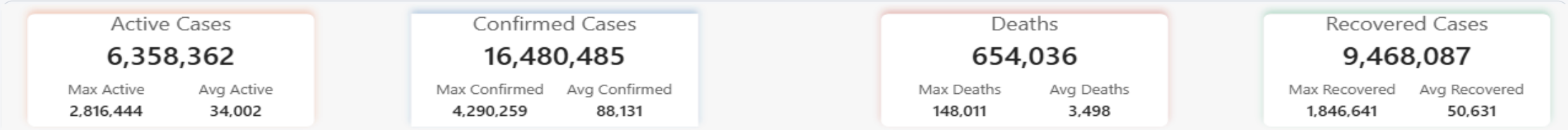
Page 2 · Category Analysis

Part 2: one category at a time, chosen with the buttons

Built in Power BI from the cleaned Python export · colours are fixed per category across the report

CHART 1 · OVERVIEW PAGE

KPI cards: totals, maximum, average



What it shows

One card per category with the world total, the highest single-country value (Max) and the per-country average (Avg). It answers Part 1 at a glance.

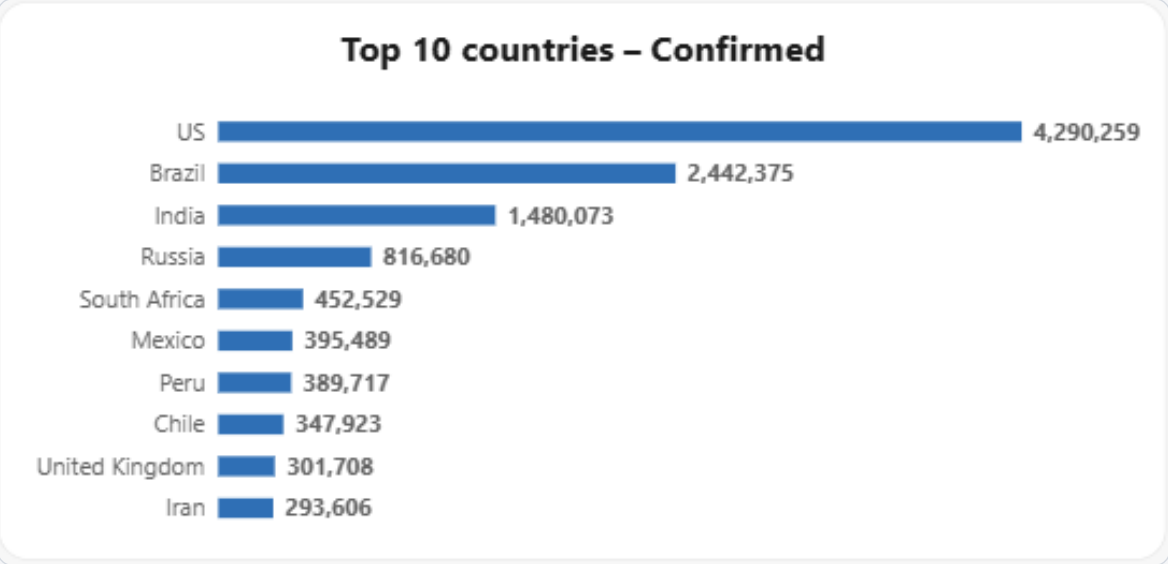
What we learn

- Of 16.5M confirmed cases, 57.5% recovered, 38.6% are still active and 4.0% died.
- The top country dwarfs the average: US confirmed is 49× the average, US active 83×, US deaths 42×, Brazil recovered 37×.

Source: Power BI dashboard, Overview page · rates computed from the world totals

CHART 2 · OVERVIEW PAGE

Top 10 countries by confirmed cases



What it shows

The ten countries with the most confirmed cases, ranked from the largest.

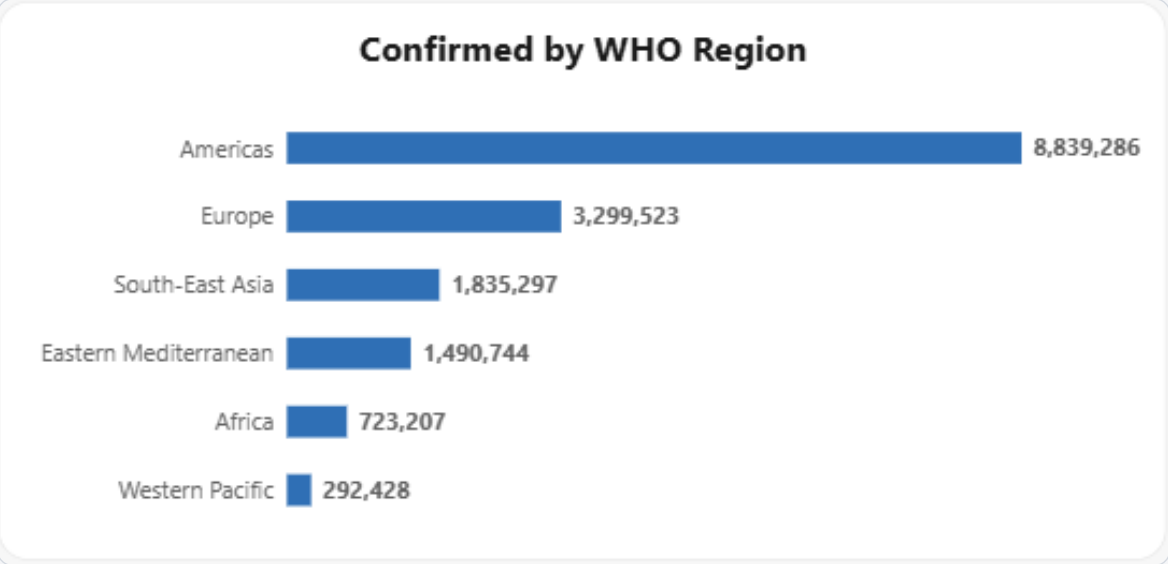
What we learn

- The US (4.29M) and Brazil (2.44M) alone are 40.9% of all confirmed cases.
- The ten countries together hold 68.0% of the world total.
- Even the tenth, Iran with 293,606, is 3.3× the average.

Source: Power BI dashboard, Overview page · shares computed from the world total of 16,480,485

CHART 3 · OVERVIEW PAGE

Confirmed cases by WHO region



What it shows

Total confirmed cases for each of the six WHO regions.

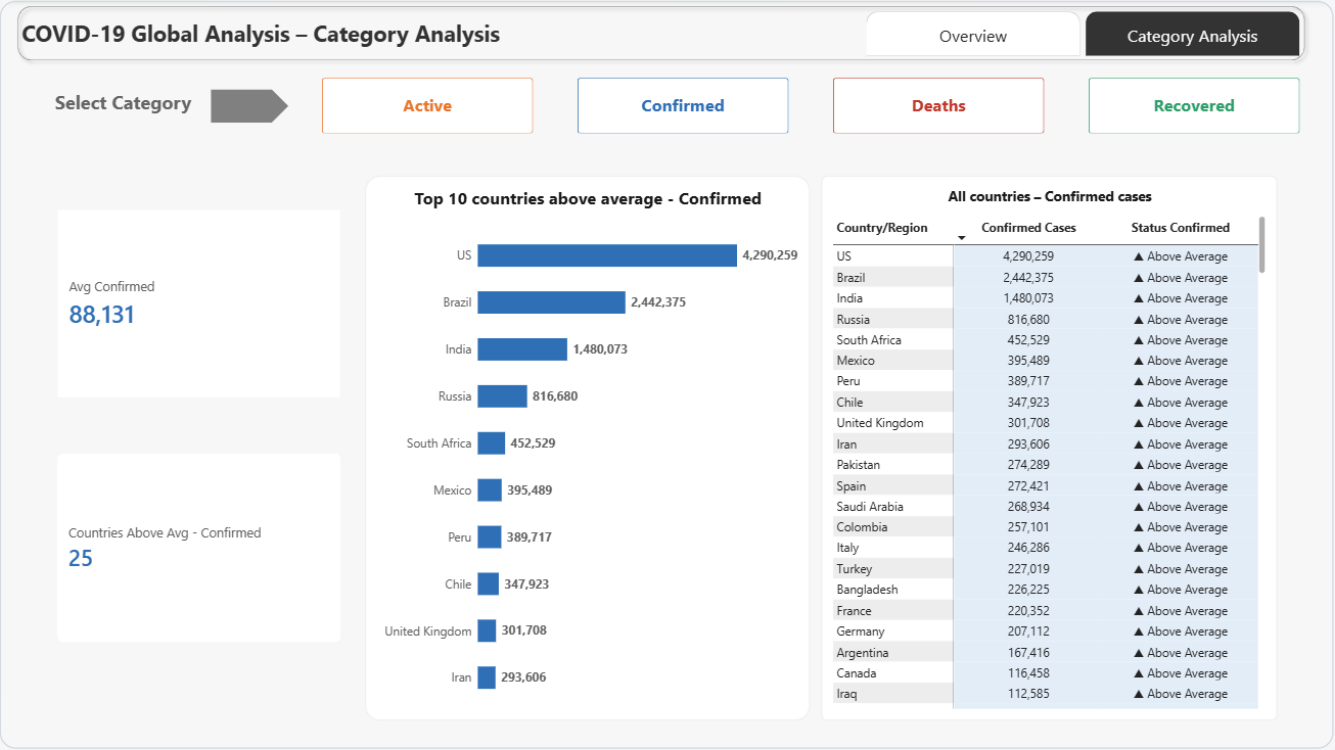
What we learn

- The Americas hold 8.84M cases, 53.6% of the world total.
- With Europe (3.30M, 20.0%) they make up 73.7% of all cases.
- The Western Pacific is smallest: 292,428 cases, 1.8%.
- These are raw counts, not adjusted for population.

Source: Power BI dashboard, Overview page · shares computed from the world total of 16,480,485

CHARTS 4 TO 7 · CATEGORY ANALYSIS PAGE

How the category page works



1 · Choose a category

Four buttons switch every visual: Active, Confirmed, Deaths, Recovered.

2 · Read the cards

The category average and how many countries are above it.

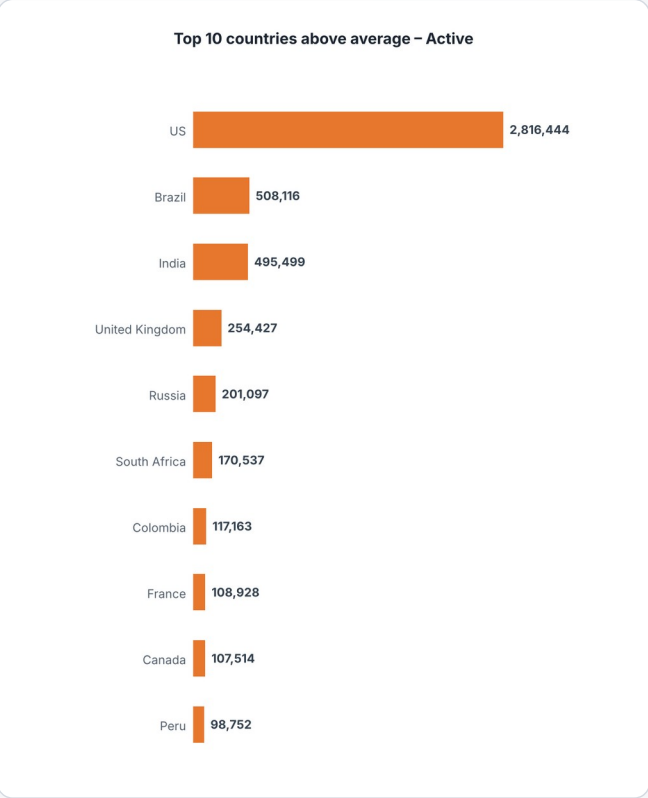
3 · Rank and check

The bar chart ranks the leading countries above average. The table lists all 187 countries with a status, and above-average rows are shaded.

Source: Power BI dashboard, Category Analysis page (Confirmed view shown)

CHART 4 · CATEGORY ANALYSIS · ACTIVE

Active: the US holds 44% of cases



All countries – Active cases

Country/Region	Active Cases	Status Active
US	2,816,444	▲ Above Average
Brazil	508,116	▲ Above Average
India	495,499	▲ Above Average
United Kingdom	254,427	▲ Above Average
Russia	201,097	▲ Above Average
South Africa	170,537	▲ Above Average
Colombia	117,163	▲ Above Average
France	108,928	▲ Above Average
Canada	107,514	▲ Above Average
Peru	98,752	▲ Above Average
Bangladesh	97,577	▲ Above Average
Spain	93,613	▲ Above Average
Argentina	91,782	▲ Above Average
Sweden	73,695	▲ Above Average
Philippines	53,649	▲ Above Average
Egypt	52,992	▲ Above Average
Mexico	47,657	▲ Above Average
Netherlands	47,064	▲ Above Average
Bolivia	47,056	▲ Above Average
Saudi Arabia	43,238	▲ Above Average
Ecuador	40,733	▲ Above Average
Belgium	39,154	▲ Above Average

What it shows

The top 10 countries above the 34,002 average, and the table of all 187 countries with their status.

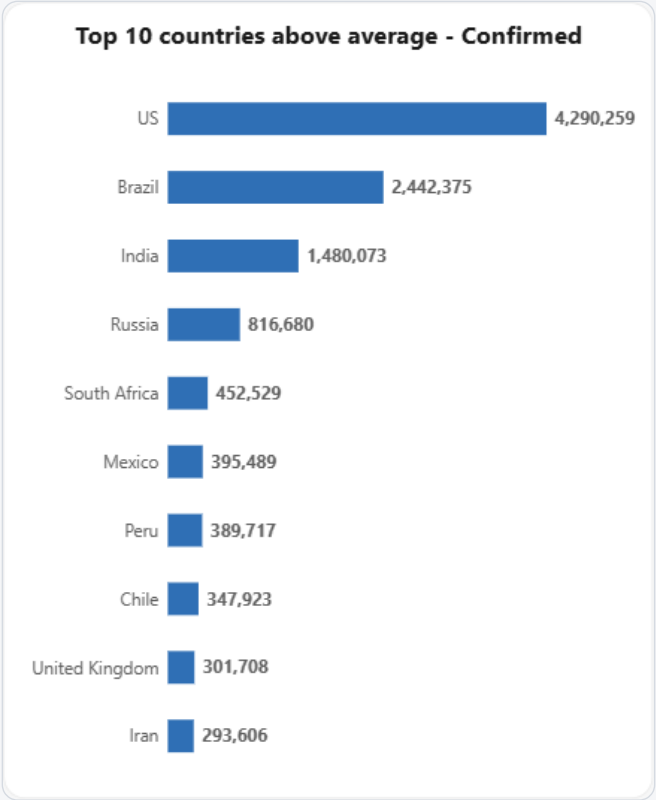
What we learn

- US: 2.82M active cases, 44.3% of the world total.
- The top five countries hold 67.2% of all active cases.
- The median is only 1,600, far below the average.
- UK and Canada report almost no recoveries, which inflates their active counts.

Source: Power BI dashboard, Category Analysis page · shares computed from the world active total of 6,358,362

CHART 5 · CATEGORY ANALYSIS · CONFIRMED

Confirmed: 25 countries above 88,131



All countries – Confirmed cases

Country/Region	Confirmed Cases	Status Confirmed
US	4,290,259	▲ Above Average
Brazil	2,442,375	▲ Above Average
India	1,480,073	▲ Above Average
Russia	816,680	▲ Above Average
South Africa	452,529	▲ Above Average
Mexico	395,489	▲ Above Average
Peru	389,717	▲ Above Average
Chile	347,923	▲ Above Average
United Kingdom	301,708	▲ Above Average
Iran	293,606	▲ Above Average
Pakistan	274,289	▲ Above Average
Spain	272,421	▲ Above Average
Saudi Arabia	268,934	▲ Above Average
Colombia	257,101	▲ Above Average
Italy	246,286	▲ Above Average
Turkey	227,019	▲ Above Average
Bangladesh	226,225	▲ Above Average
France	220,352	▲ Above Average
Germany	207,112	▲ Above Average
Argentina	167,416	▲ Above Average
Canada	116,458	▲ Above Average
Iraq	112,585	▲ Above Average

What it shows

The top 10 countries above the 88,131 average, and the table of all 187 countries with their status.

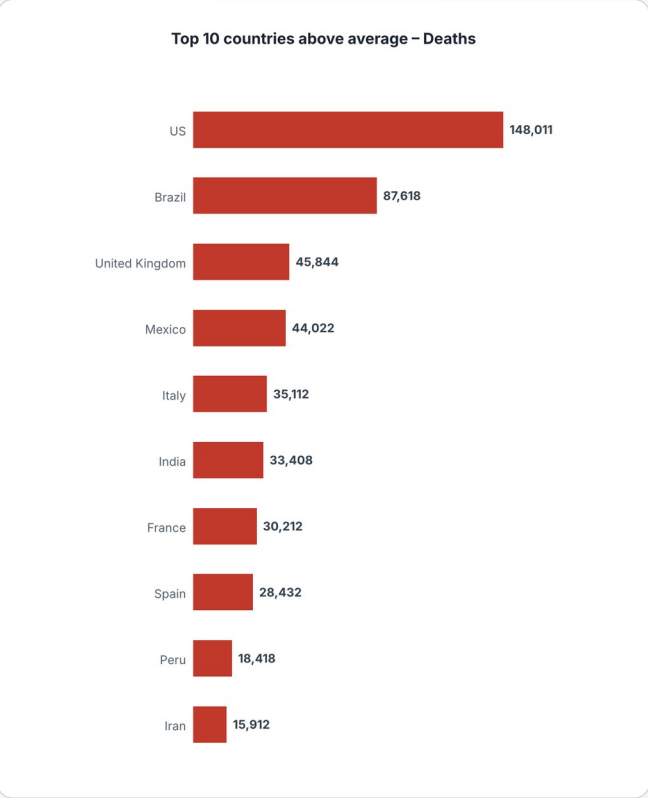
What we learn

- 25 countries hold 85.6% of all confirmed cases.
- US: 4.29M cases, 26.0% of the world; Brazil follows with 2.44M.
- The top five countries hold 57.5% of all cases.
- The median is 5,059, so the average is 17× a typical country.

Source: Power BI dashboard, Category Analysis page · shares computed from the world confirmed total of 16,480,485

CHART 6 · CATEGORY ANALYSIS · DEATHS

Deaths: 26 countries hold 91.8%



All countries – Deaths

Country/Region	Deaths	Status Deaths
US	148,011	▲ Above Average
Brazil	87,618	▲ Above Average
United Kingdom	45,844	▲ Above Average
Mexico	44,022	▲ Above Average
Italy	35,112	▲ Above Average
India	33,408	▲ Above Average
France	30,212	▲ Above Average
Spain	28,432	▲ Above Average
Peru	18,418	▲ Above Average
Iran	15,912	▲ Above Average
Russia	13,334	▲ Above Average
Belgium	9,822	▲ Above Average
Chile	9,187	▲ Above Average
Germany	9,125	▲ Above Average
Canada	8,944	▲ Above Average
Colombia	8,777	▲ Above Average
South Africa	7,067	▲ Above Average
Netherlands	6,160	▲ Above Average
Pakistan	5,842	▲ Above Average
Sweden	5,700	▲ Above Average
Turkey	5,630	▲ Above Average
Ecuador	5,532	▲ Above Average

What it shows

The top 10 countries above the 3,498 average, and the table of all 187 countries with their status.

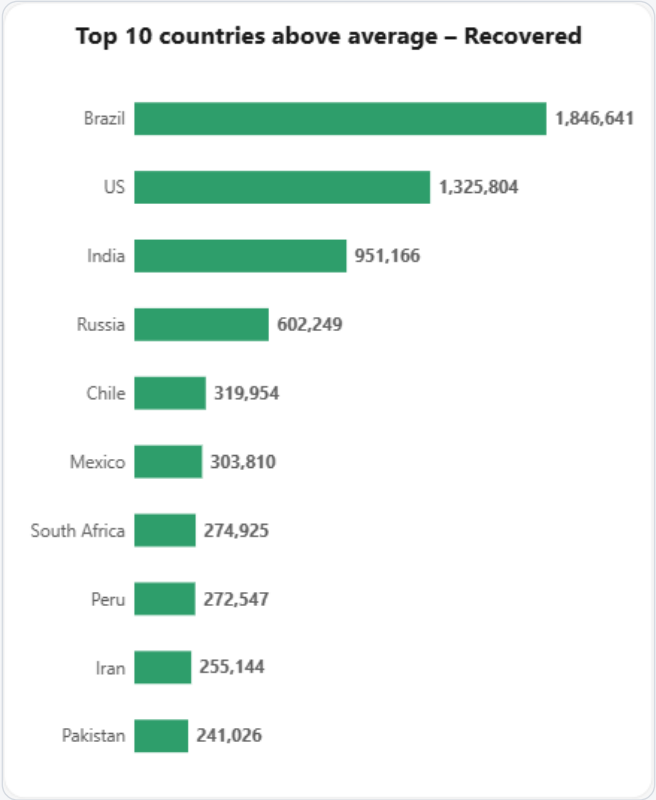
What we learn

- US: 148,011 deaths (22.6%), then Brazil 87,618 and the UK 45,844.
- The top five countries hold 55.1% of all deaths.
- 17 countries report zero deaths; the median is 108.
- The UK (301,708 cases) has more deaths than India (1,480,073 cases).

Source: Power BI dashboard, Category Analysis page · shares computed from the world deaths total of 654,036

CHART 7 · CATEGORY ANALYSIS · RECOVERED

Recovered: Brazil leads, ahead of the US



All countries – Recovered cases

Country/Region	Recovered Cases	Status Recovered
Brazil	1,846,641	▲ Above Average
US	1,325,804	▲ Above Average
India	951,166	▲ Above Average
Russia	602,249	▲ Above Average
Chile	319,954	▲ Above Average
Mexico	303,810	▲ Above Average
South Africa	274,925	▲ Above Average
Peru	272,547	▲ Above Average
Iran	255,144	▲ Above Average
Pakistan	241,026	▲ Above Average
Saudi Arabia	222,936	▲ Above Average
Turkey	210,469	▲ Above Average
Italy	198,593	▲ Above Average
Germany	190,314	▲ Above Average
Spain	150,376	▲ Above Average
Colombia	131,161	▲ Above Average
Bangladesh	125,683	▲ Above Average
Qatar	106,328	▲ Above Average
France	81,212	▲ Above Average
China	78,869	▲ Above Average
Iraq	77,144	▲ Above Average
Argentina	72,575	▲ Above Average

What it shows

The top 10 countries above the 50,631 average, and the table of all 187 countries with their status.

What we learn

- Brazil (1.85M) is first and the US (1.33M) second, the only category the US does not lead.
- 28 countries hold 88.5% of all recoveries; the top five hold 53.3%.
- Six countries report zero recoveries, including Canada, Sweden and Serbia.

Source: Power BI dashboard, Category Analysis page · shares computed from the world recovered total of 9,468,087

INSIGHT 1 · CONCENTRATION

A few countries carry most cases

Active

89.3%

of active cases sit in 24 countries (12.8%)

Confirmed

85.6%

of confirmed cases sit in 25 countries (13.4%)

Deaths

91.8%

of deaths sit in 26 countries (13.9%)

Recovered

88.5%

of recoveries sit in 28 countries (15.0%)

About one country in seven carries 86% to 92% of every category

The top five countries alone hold 67.2% of Active, 57.5% of Confirmed, 55.1% of Deaths and 53.3% of Recovered.

Source: computed from country_wise_latest · share of each world total held by the countries above that category's average

INSIGHT 2 · AVERAGE VS MEDIAN

The average is far above a typical country

Category	Average	Median	Average ÷ median
Active	34,002	1,600	21×
Confirmed	88,131	5,059	17×
Deaths	3,498	108	32×
Recovered	50,631	2,815	18×

Why it matters

A handful of very large outbreaks pull the mean up. So “above average” isolates the heavy hitters, 13% to 15% of countries, not simply the countries with a problem. Reporting the median next to the mean shows the typical country.

Source: computed from country_wise_latest · 187 countries · snapshot of 27 Jul 2020

INSIGHT 3 · REGIONS

Regions and outcome rates

WHO region	Share of confirmed	Deaths per 100 cases	Recovered per 100 cases
Americas	53.6%	3.9	50.6
Europe	20.0%	6.4	60.4
South-East Asia	11.1%	2.3	63.0
Eastern Mediterranean	9.0%	2.6	80.6
Africa	4.4%	1.7	60.9
Western Pacific	1.8%	2.8	70.7
World	100%	4.0	57.5

What we learn

- Europe has the highest death rate: 6.4 per 100 cases against 4.0 worldwide.
- The UK, Belgium, Italy and France are all above 13 deaths per 100 cases.
- Eastern Mediterranean has the highest recovery rate: 80.6 per 100 cases.
- Rates use reported cases, and testing differs by country.

Source: computed from country_wise_latest · country rates for countries with 10,000+ confirmed cases

INSIGHT 4 · CAVEATS

Data flags that change the reading

Recoveries are under-reported

Canada, Serbia and Sweden report zero recovered; the UK (1,437) and the Netherlands (189) report almost none. Active = Confirmed − Deaths − Recovered, so their active counts are overstated. UK active of 254,427 ranks fourth.

Overlap across categories

11 countries are above average in all four categories: Brazil, Colombia, France, India, Indonesia, Mexico, Peru, Russia, South Africa, Spain and the US. 38 countries are above average in at least one.

Counts, not rates

The data has no population or testing figures, so large countries rank high by size alone. It is also one snapshot, 27 July 2020, with no trend inside the country table.

Source: computed from country_wise_latest · 187 countries · snapshot of 27 Jul 2020

RECOMMENDATIONS · 1 OF 2

Prioritizing Public Health Response

1 · Prioritize healthcare resource allocation

Assess healthcare resource needs in countries with the highest case burden, including medical supplies and hospital capacity.

2 · Strengthen Mortality Reduction Efforts

Identify opportunities to improve critical care capacity and timely access to treatment in high-mortality countries.

3 · Develop Targeted Regional Response Strategies

Prioritize the Americas and Europe, which account for 73.7% of confirmed cases, through region-specific response strategies.

4 · Investigate Regional Differences in Mortality

Examine potential factors contributing to Europe's higher reported death rate of 6.4 per 100 confirmed cases, compared with 4.0 globally.

Evidence: concentration, average vs median and regional slides

RECOMMENDATIONS · 2 OF 2

Supporting Public Health Decision-Making

5 · Strengthen Healthcare System Preparedness

Assess hospital capacity, ICU availability, medical staffing, and essential supplies in high-burden countries.

6 · Strengthen Surveillance and Early Detection

Improve testing, case reporting, and epidemiological surveillance to support timely outbreak detection and response..

7 · Enhance Access to Critical Care

Evaluate access to timely medical care and critical care services in countries with high reported mortality.

8 · Establish Continuous Monitoring

Track key public health indicators over time to identify changes and evaluate response effectiveness.

9 · Support Evidence-Based Country-Level Interventions

Combine case data with population and healthcare information to develop targeted interventions based on each country's needs.

Evidence: concentration, average vs median and regional slides